2. Curves, Surfaces and Vector Fields

An overview

- A parametrized curve is a differentiable map $\mathbf{c} : \mathbb{R} \to \mathbb{R}^3$.
- A parametrized surface is a differentiable map $\Phi : \mathbb{R}^2 \to \mathbb{R}^3$.
- A vector field is a function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$.
- A curvilinear coordinate is a differentiable function $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$

$$\begin{aligned} x &= f_1(u_1, u_2, u_3) \\ y &= f_2(u_1, u_2, u_3) \\ z &= f_3(u_1, u_2, u_3). \end{aligned}$$

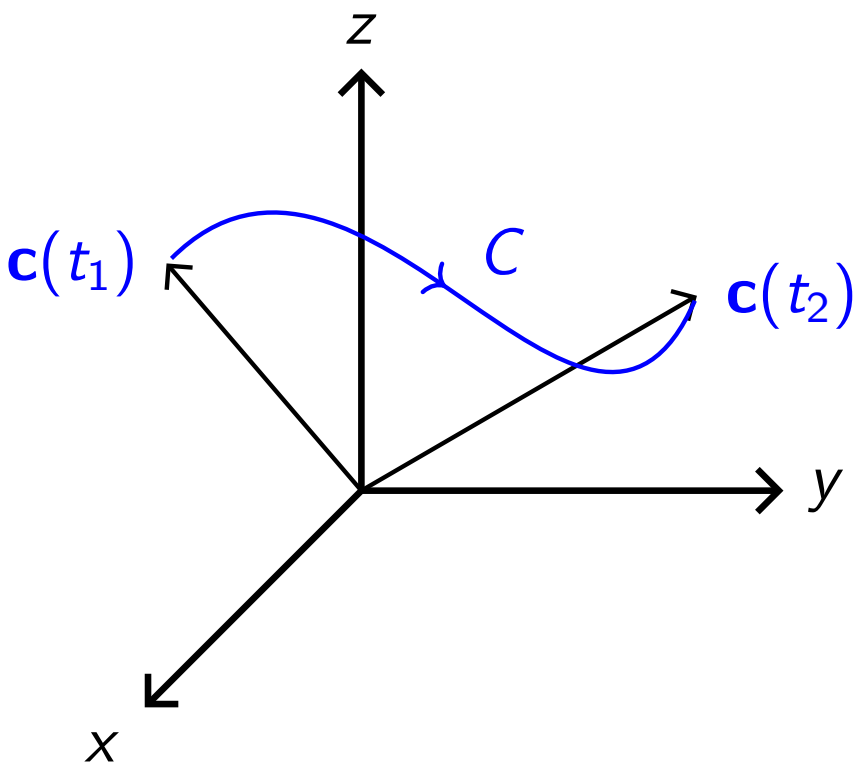
such that $\mathbf{DF} \neq 0$.

2.9. Parametric paths

The position at time t of a particle moving in space is given by the path $\mathbf{c}$ where

$$\mathbf{c}(t) = (x(t), y(t), z(t)) \,.$$

$\mathbf{c}$ parametrises the curve C, which is traced out by $\mathbf{c}(t)$ as t varies.

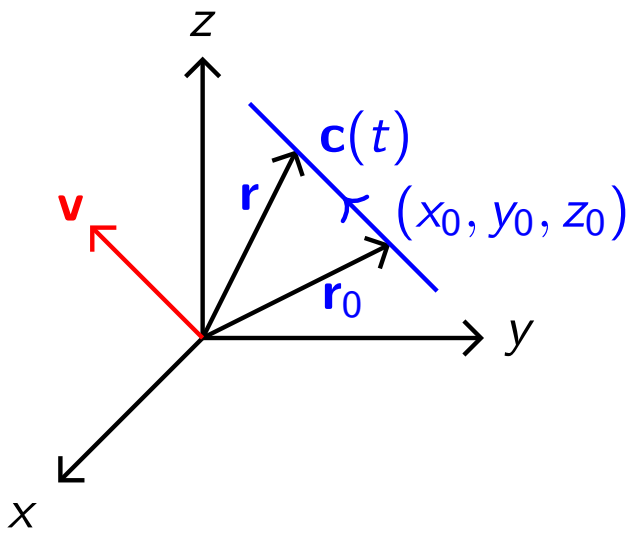


An orientation of a curve is a choice of direction of the curve.

A parametrization is said to be orientation preserving if the parameterization with t increasing goes in the direction of the orientation.

Example 9.1

Determine the parametrisation of a line through (x_0, y_0, z_0) in the direction of a vector $\mathbf{v}$.



Example 9.2

Find a parametrisation for the following curve with t increasing. The line segment joining $(1, 0, 2)$ to $(4, 3, 1)$

Example 9.3

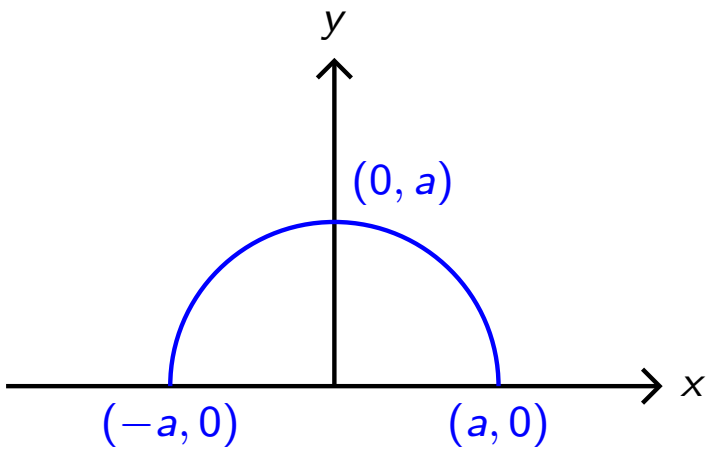
What kind of curve is given by $\mathbf{c}(t) = (\cos t, \sin t, t)$?

Example 9.4

Find an orientation preserving parametrisation for the following curve.
The semicircle $y = \sqrt{a^2 - x^2}$ oriented

(i) anticlockwise

(ii) clockwise.



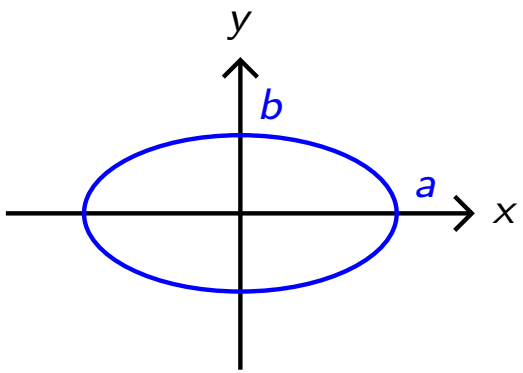
Example 9.5

Find an orientation preserving parametrisation for the following curve.
The ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(i) anticlockwise

(ii) clockwise



Example 9.6

Find a parametrisation for the following curves. The parabola

$$y = z^2, \ x = 2$$

from

(i) $(2, 0, 0)$ to $(2, 9, 3)$ and

(ii) $(2, 9, 3)$ to $(2, 0, 0)$.

Path Properties

Let $\mathbf{c} : \mathbb{R} \to \mathbb{R}^3$ be a differentiable path in $\mathbb{R}^3$, with $t \mapsto (x(t), y(t), z(t))$.

1. velocity:

$$\mathbf{v}(t) = \frac{d\mathbf{c}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$$

   speed:

$$|\mathbf{v}(t)| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.$$

   The direction of $\mathbf{v}(t)$ is tangent to the path.

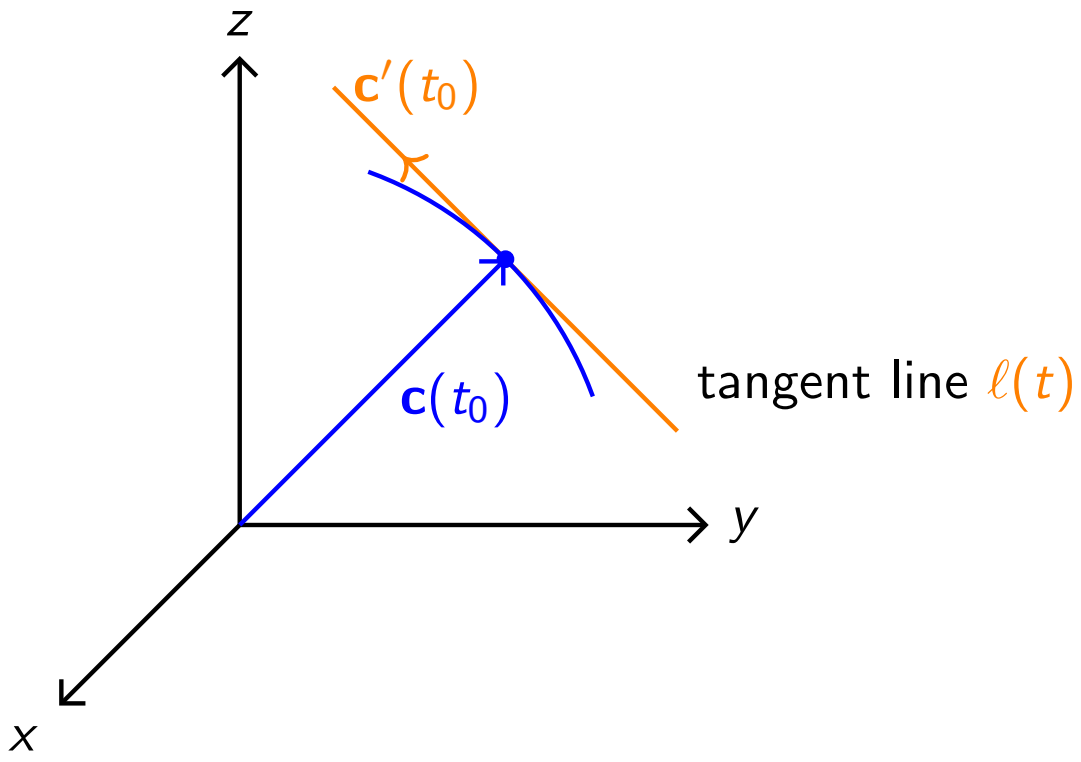
2. acceleration:

$$\mathbf{a}(t) = \frac{d^2\mathbf{c}}{dt^2} = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2}\right)$$

3. tangent line to $\mathbf{c}$ at t_0:

$$\ell(t) \;=\; \mathbf{c}(t_0) + (t - t_0)\mathbf{c}'(t_0)$$

for $t \in \mathbb{R}$.



Example 9.7

Let $\mathbf{c}(t) = (3t + 2, t^2 - 7, t - t^2)$. Determine the velocity, speed, acceleration and equation of the tangent line at $t = 1$.

Differentiation Rules

Let $\mathbf{b}$, $\mathbf{c}$ be differentiable paths in $\mathbb{R}^3$. Then

1. $\dfrac{d}{dt}[\mathbf{b}+\mathbf{c}] = \dfrac{d\mathbf{b}}{dt} + \dfrac{d\mathbf{c}}{dt}$
2. $\dfrac{d}{dt}[\mathbf{b}\cdot\mathbf{c}] = \dfrac{d\mathbf{b}}{dt}\cdot\mathbf{c} + \mathbf{b}\cdot\dfrac{d\mathbf{c}}{dt}$
3. $\dfrac{d}{dt}[\mathbf{b}\times\mathbf{c}] = \dfrac{d\mathbf{b}}{dt}\times\mathbf{c} + \mathbf{b}\times\dfrac{d\mathbf{c}}{dt}$

Example 9.8

Prove differentiation rule 2.

Example 9.9

Let $\mathbf{u} : \mathbb{R} \to \mathbb{R}^3$, and suppose $\mathbf{u}'''$ exists. Evaluate and simplify

$$\frac{d}{dt}\left[(\mathbf{u}'' \times \mathbf{u}') \cdot \mathbf{u}\right].$$

2.10. Tangent vectors and normal vectors

$\mathbf{T}(t)$ is the unit tangent vector to the path $\mathbf{c}$ at the point $\mathbf{c}(t)$.

$$\mathbf{T}(t) = \frac{\dfrac{d\mathbf{c}}{dt}}{\left|\dfrac{d\mathbf{c}}{dt}\right|}$$

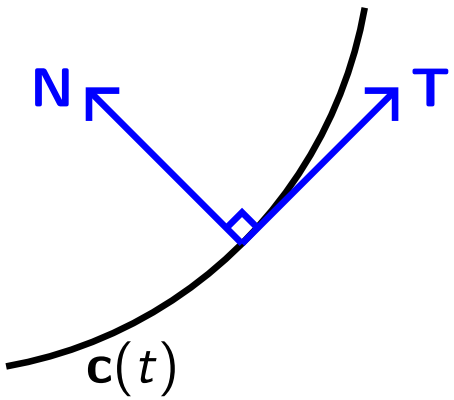
$$\text{speed} = \left|\frac{d\mathbf{c}}{dt}\right|$$

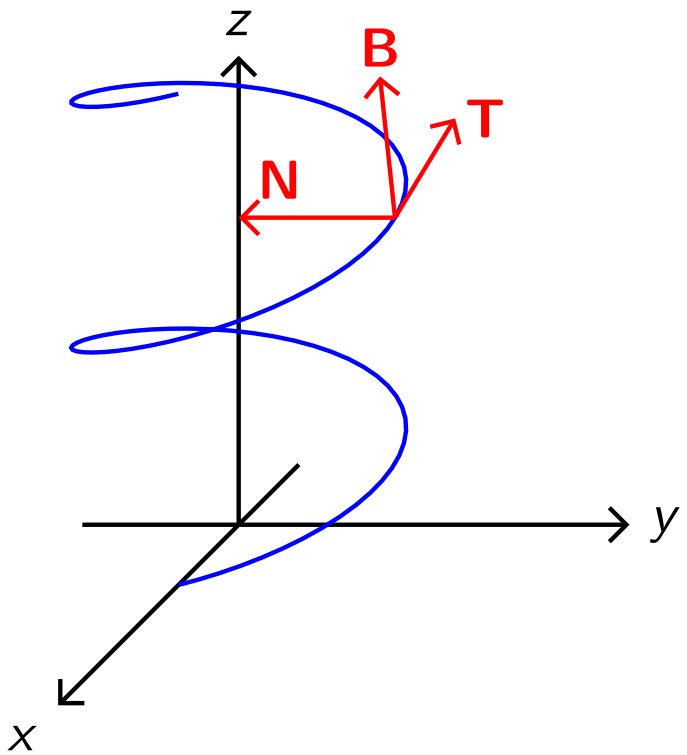
Normal vectors to a path

The principal normal vector

$$\mathbf{N}(t) = \frac{\dfrac{d\mathbf{T}}{dt}}{\left|\dfrac{d\mathbf{T}}{dt}\right|}$$

is a unit normal vector to the path $\mathbf{c}$ at the point $\mathbf{c}(t)$.





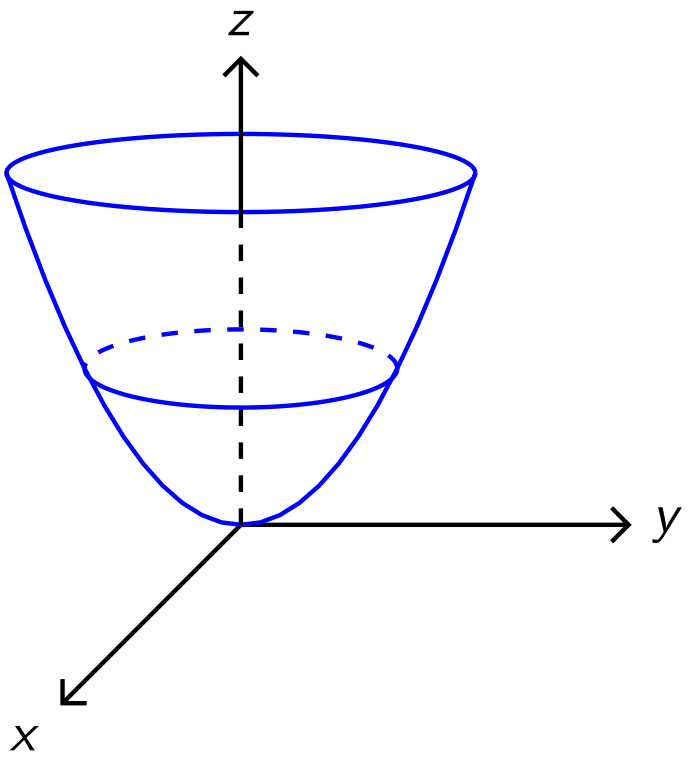
A third unit vector which is perpendicular to both **T** and **N** is the binormal vector **B**:

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

2.11. Parametrised surfaces

Some surfaces have the form $z = f(x, y)$.

Example: $z = x^2 + y^2$

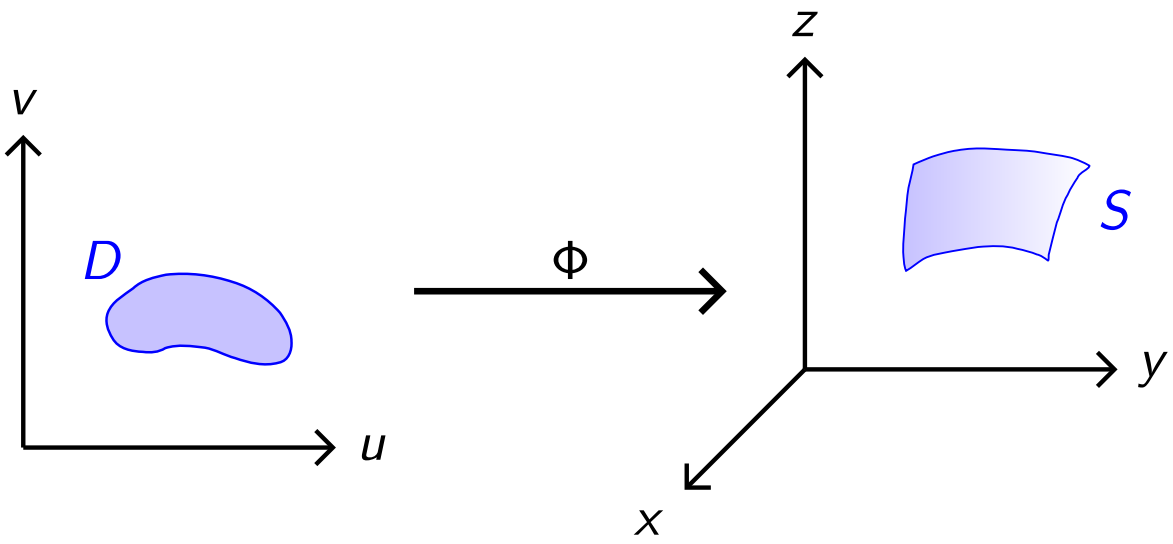


Some surfaces satisfy $f(x, y, z) = c$. For example

- $x - z + z^3 = 0$
- torus

A parametrised surface is a function Φ that maps a domain D in $\mathbb{R}^2$ to a surface S in $\mathbb{R}^3$, namely

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)) .$$



Note

If Φ is C^1, then S is said to be a differentiable or C^1 surface.

Example 11.1

Let

$$\Phi : x = u\cos v,\ \ y = u\sin v,\ \ z = u$$

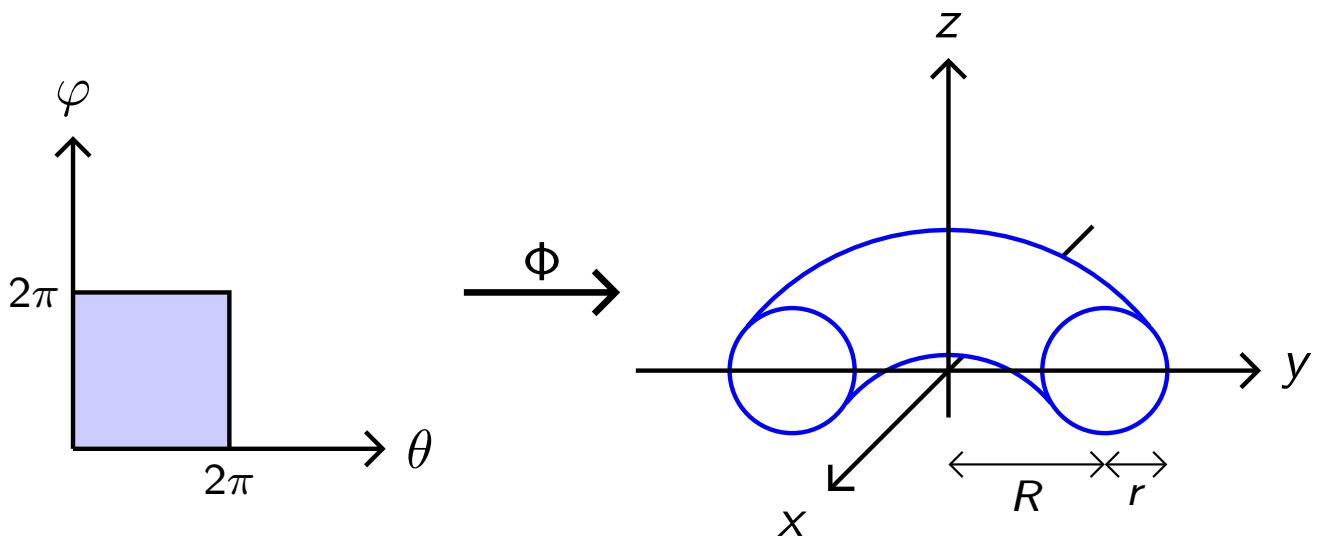
for $0 \le v \le 2\pi,\ \ u \ge 0$. Identify the surface.

Example 11.2

A torus is parametrised by

$$\begin{aligned}\Phi : x &= (R + r\cos\varphi)\cos\theta \\ y &= (R + r\cos\varphi)\sin\theta \\ z &= r\sin\varphi,\end{aligned}$$

for $0 \leq \theta \leq 2\pi$ and $0 \leq \varphi \leq 2\pi$. Here R and r are fixed real numbers with $R > r$.



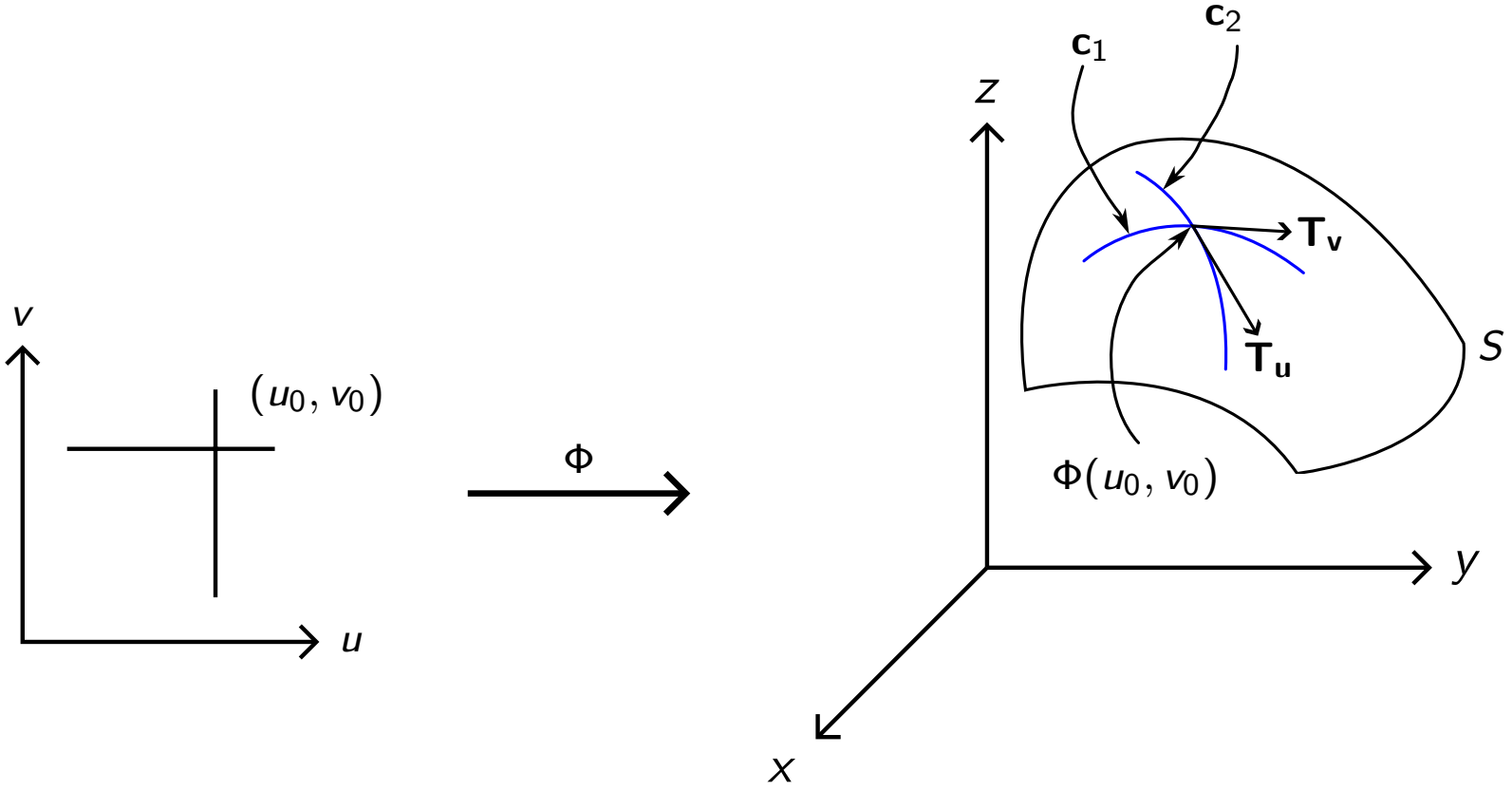
Example 11.3

Parametrise the surface of the parabolic bowl $z = x^2 + y^2$.

Tangents and Normals to Surfaces

Let S be a differentiable surface. Consider curves on S given by

$$\mathbf{c}_1(v) = \Phi(u_0, v), \quad (u = u_0 \text{ is constant})$$
$$\mathbf{c}_2(u) = \Phi(u, v_0), \quad (v = v_0 \text{ is constant}).$$



- $\mathbf{T_v}(v_0)$ is the tangent vector to $\mathbf{c}_1$ at $\Phi(u_0, v_0)$

$$\mathbf{T_v}(v_0) = \left.\frac{d\mathbf{c}_1}{dv}\right|_{v=v_0} = \left.\left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)\right|_{(u_0,v_0)}$$

- $\mathbf{T_u}(u_0)$ is the tangent vector to $\mathbf{c}_2$ at $\Phi(u_0, v_0)$

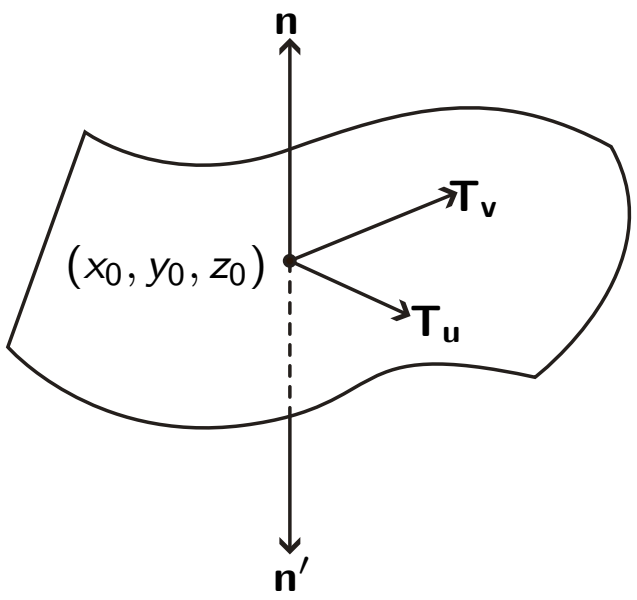
$$\mathbf{T_u}(u_0) = \left.\frac{d\mathbf{c}_2}{du}\right|_{u=u_0} = \left.\left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)\right|_{(u_0,v_0)}$$

Note

$\mathbf{T_u}$, $\mathbf{T_v}$ change with position (x, y, z). They need not be orthogonal to each other.

- There are two normal vectors to the surface S at (x_0, y_0, z_0) pointing in opposite directions:

$$\mathbf{n} = \mathbf{T_u} \times \mathbf{T_v}$$
$$\mathbf{n}' = -\mathbf{n} = \mathbf{T_v} \times \mathbf{T_u}.$$



Note

1. The normal vector $\mathbf{n}$ changes with position $(x, y, z) = \phi(u, v)$.
2. If $\mathbf{n}(u_0, v_0) \neq \mathbf{0}$, then the parameterization is said to be smooth at the point $\phi(u_0, v_0)$.

Example 11.4

Find the normal vector to the cone parametrised by

$$\Phi : x = u\cos v,\ \ y = u\sin v,\ \ z = u$$

for

$$0 \leq v \leq 2\pi,\ \ u \geq 0$$

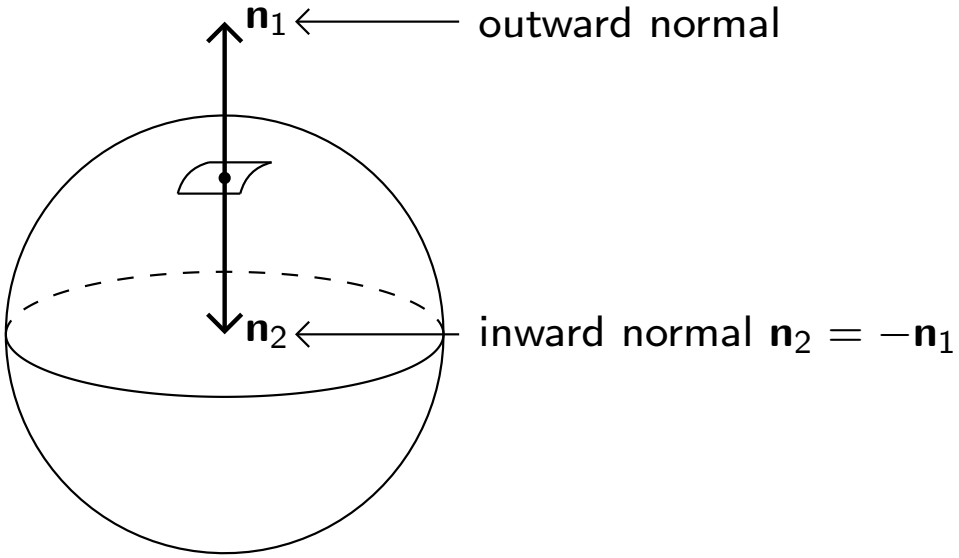
Orientable surfaces

A surface S is orientable if S has a unit normal vector field $\hat{\mathbf{n}}$ that varies continuously over S. We will only consider orientable surfaces.

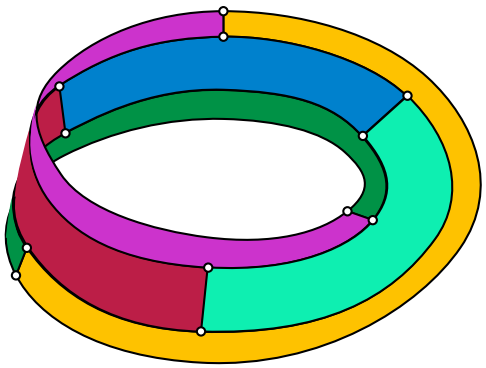
Intuitively, an orientable surface has two sides.

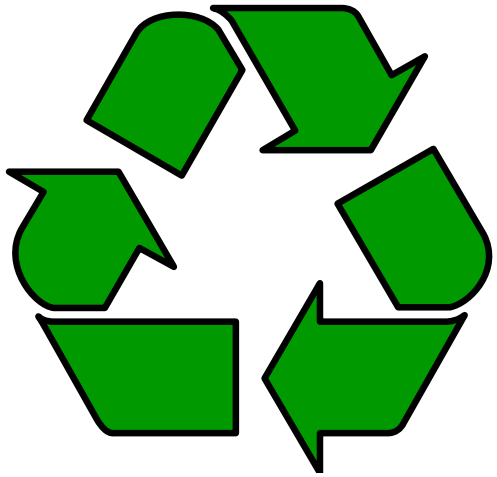
An orientation on an orientable surface S is a choice of a unit normal vector field $\hat{\mathbf{n}}$ (there are two possible choices). We say that S is oriented if an orientation is given.

Example 1: A sphere is an orientable surface.



Example 2: A Möbius band is not an orientable surface. At each point there are 2 normal directions but as it is possible to travel along the surface from one side to the other, there is no consistent choice of unit normal.





Tangent Plane to Parametrised Surfaces

If S is a parametric surface that is smooth at $\phi(u_0, v_0) = (x_0, y_0, z_0)$, then the Cartesian equation of the tangent plane to S at (x_0, y_0, z_0) is

$$(x - x_0, y - y_0, z - z_0) \cdot \mathbf{n}(u_0, v_0) = 0$$

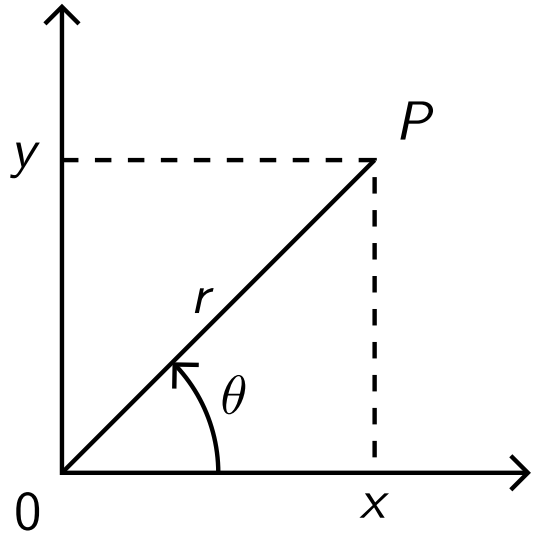
where $\mathbf{n}(u_0, v_0)$ is normal to S at (x_0, y_0, z_0).

Example 11.5

Find the Cartesian equation of the tangent plane to the cone in Example (11.4) at $(1, 1, \sqrt{2})$.

Polar Coordinates (r, θ)

We can specify a point in the xy plane by Cartesian (x, y) or polar (r, θ) coordinates.



- $r = \left|\overrightarrow{OP}\right| = \sqrt{x^2 + y^2}$. We have $r \geq 0$.
- $x = r\cos\theta$ and $y = r\sin\theta$.
- $\theta =$ angle measured anticlockwise from positive x axis to $\overrightarrow{OP}$.

Note

1. The origin is given by $(0, \theta)$ where θ can be any real number.
2. θ is not unique, e.g. $(x, y) = (1, 1)$ has polar coordinates $\left(\sqrt{2}, \frac{\pi}{4}\right)$, $\left(\sqrt{2}, \frac{9\pi}{4}\right)$, $\left(\sqrt{2}, \frac{-7\pi}{4}\right)$ etc.
3. To find $\theta(x, y)$ we restrict θ to $0 \leq \theta < 2\pi$. Then

$$\theta = \begin{cases} \arctan\left(\frac{y}{x}\right), & \text{if } x > 0,\ y \geq 0, \\ \pi + \arctan\left(\frac{y}{x}\right) & \text{if } x < 0, \\ 2\pi + \arctan\left(\frac{y}{x}\right) & \text{if } x > 0,\ y < 0, \\ \frac{\pi}{2} & \text{if } x = 0,\ y > 0, \\ \frac{3\pi}{2} & \text{if } x = 0,\ y < 0. \end{cases}$$

Note that $-\dfrac{\pi}{2} < \arctan\left(\dfrac{y}{x}\right) < \dfrac{\pi}{2}$.

Example 12.1

Convert $(x, y) = (-1, -1)$ to polar coordinates.

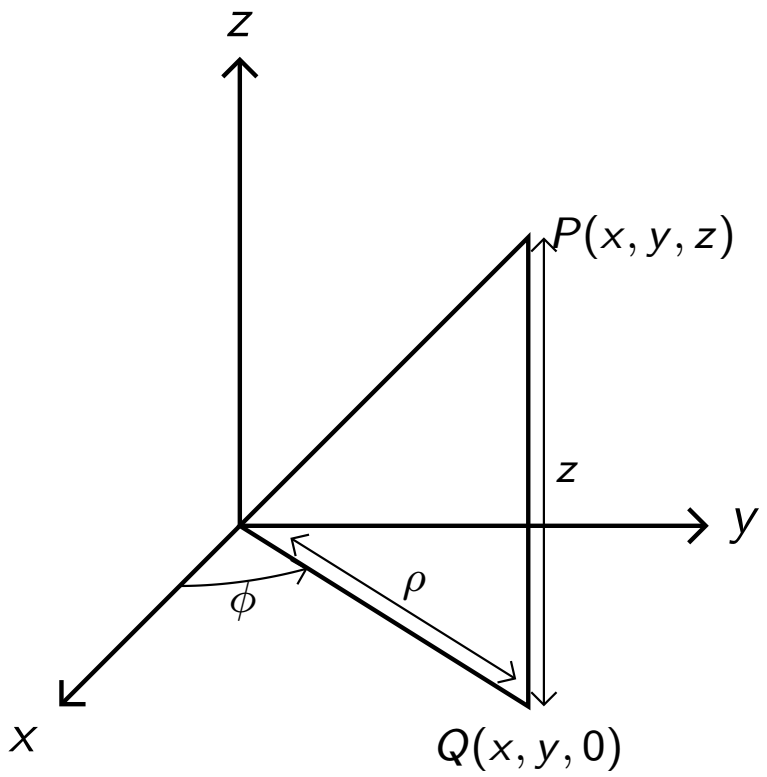
Special Cases

$\underline{r = a \text{ (constant)}}$

- $\underline{\theta = \alpha \text{ (constant)}}$

Cylindrical Coordinates (ρ, ϕ, z)

We can specify a point in 3 dimensional space by Cartesian (x, y, z), cylindrical (ρ, ϕ, z) or spherical (r, θ, ϕ) coordinates.



- $\rho = |\overrightarrow{OQ}| = \sqrt{x^2 + y^2}$. We have $\rho \geq 0$.
- $\phi =$ angle measured anticlockwise from the positive x axis to $\overrightarrow{OQ}$. We have

$$0 \leq \phi < 2\pi.$$

- $\phi = \arctan\left(\frac{y}{x}\right)$ if $x > 0,\ y \geq 0$. (For other x, y values, it is as in the expression for θ in polar coordinates.)
- We have

$$
\begin{aligned}
x &= \rho \cos \phi, \\
y &= \rho \sin \phi, \\
z &= z.
\end{aligned}
$$

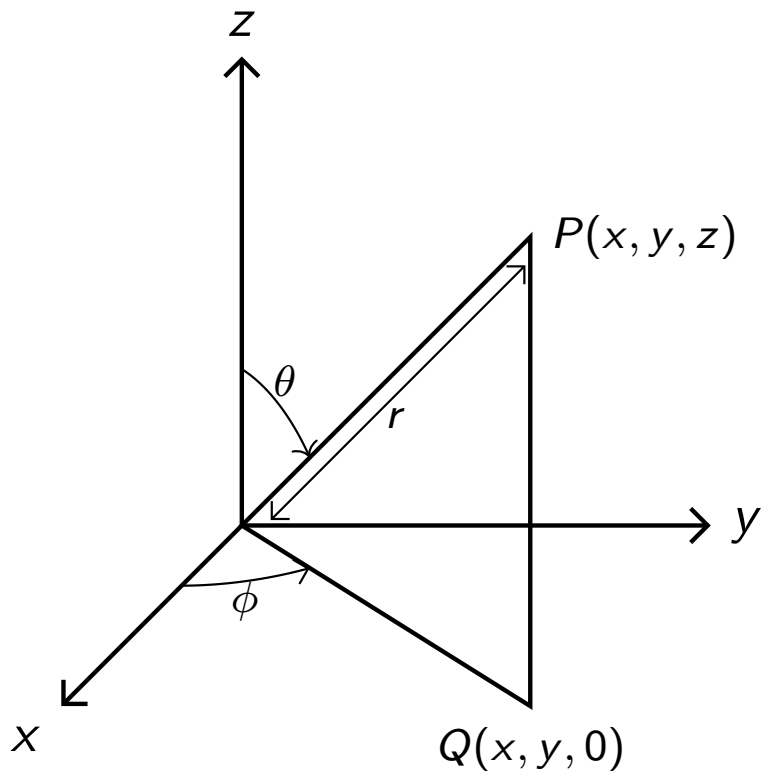
Special Cases

- $z = c$ (constant)

- $\underline{\rho = a \text{ (constant)}}$

- $\underline{\phi = \alpha \text{ (constant)}}$

Spherical Coordinates (r, θ, ϕ)



- $r =$ length of $\overrightarrow{OP} = \sqrt{x^2 + y^2 + z^2}$. We have $r \geq 0$.
- ϕ (azimuthal angle) $=$ angle measured anticlockwise from the positive x axis to $\overrightarrow{OQ}$. We have

$$0 \leq \phi < 2\pi.$$

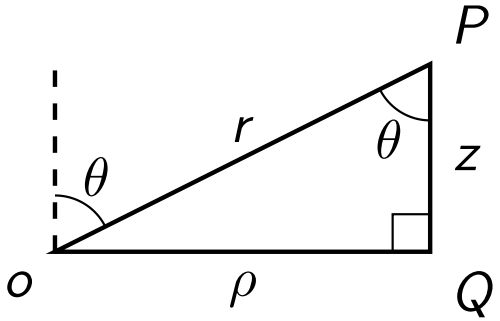
- $\phi = \arctan\left(\frac{y}{x}\right)$ if $x > 0,\ y \geq 0$. (For other x, y values, it is as in the expression for θ in polar coordinates.)
- θ (polar angle) $=$ angle measured from positive z axis to $\overrightarrow{OP}$. We have

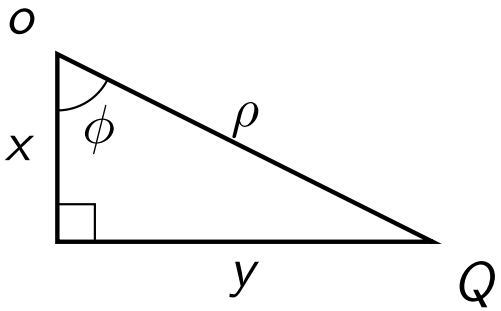
$$0 \leq \theta \leq \pi$$

$$\theta = \arccos\left(\frac{z}{r}\right).$$

- Also

$$\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta. \end{aligned}$$





$$\Rightarrow \rho = r \sin\theta \qquad \Rightarrow \begin{cases} x &= \rho\cos\phi \\ y &= \rho\sin\phi \end{cases}$$

Special Cases

- $r = a$ (constant)

- $\underline{\theta = \gamma \text{ (constant)}}$

- $\underline{\phi = \alpha \text{ (constant)}}$

Example 12.2

Convert $(x, y, z) = (1, 1, \sqrt{3})$ to cylindrical and spherical coordinates.

2.13. Vector fields

A vector field is a function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$.

$\mathbf{n = 2}$:

$$\mathbf{F}(x, y) = u(x, y)\,\mathbf{i} + v(x, y)\,\mathbf{j}$$

$\mathbf{F}(x, y)$ is a vector (u, v) where each component is a scalar function of x, y.

$\mathbf{n = 3}$:

$$\mathbf{F}(x, y, z) = u(x, y, z)\,\mathbf{i} + v(x, y, z)\,\mathbf{j} + w(x, y, z)\,\mathbf{k}$$

$\mathbf{F}(x, y, z)$ is a vector (u, v, w) where each component is a scalar function of x, y, z.

The vector field can be sketched by assigning to each point $\mathbf{x}$ a vector $\mathbf{F}(\mathbf{x})$ represented by an arrow whose tail is at $\mathbf{x}$.

A vector field $\mathbf{F}(\mathbf{x})$ can represent a physical vector quantity such as force or velocity at position $\mathbf{x}$.

Example 13.1

Sketch the vector field

$$\mathbf{F}(x, y) = x\,\mathbf{i} = (x, 0).$$

Example 13.2

Sketch the vector field

$$\mathbf{F}(x, y) = -y\,\mathbf{j} = (0, -y).$$

Example 13.3

Sketch the vector field

$$\mathbf{F}(x, y) = x\,\mathbf{i} - y\,\mathbf{j} = (x, -y).$$

Example 13.4

Sketch the vector field

$$\mathbf{F}(x, y) = -y\,\mathbf{i} + x\,\mathbf{j} = (-y, x).$$

Example 13.5

Sketch the vector field

$$\mathbf{F}(\mathbf{r}) = -\frac{GMm}{r^3}\,\mathbf{r}$$

where $\mathbf{r}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}\,,\ r = |\mathbf{r}|$.
This is the gravitational force of attraction of the earth (of mass M) on a particle (of mass m) at the position (x, y, z).

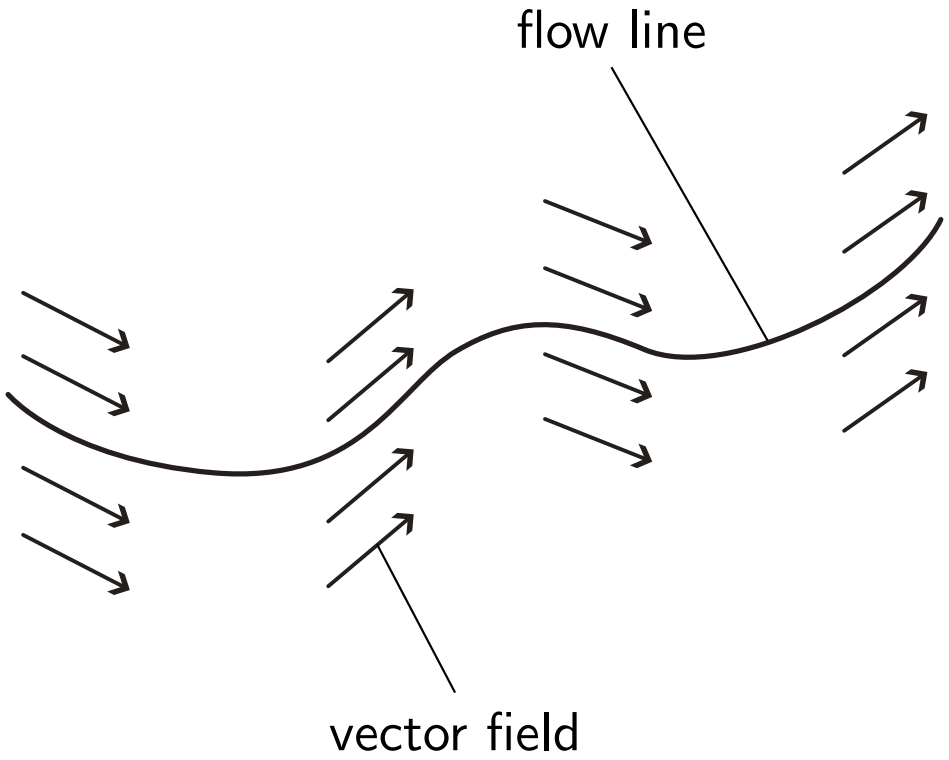
Flow Lines

A path $\mathbf{c}$ is a flow line (or streamline) of a vector field $\mathbf{F}$ if

$$\mathbf{c}'(t) \;=\; \mathbf{F}\left[\mathbf{c}(t)\right].$$

Physical Interpretation

Geometrically, a flow line is a curve whose tangent vector coincides with the vector field:



Example

Flow of water through a pipe.

Flow line represents the path followed by a particle suspended in the fluid.

Example 13.6

Show that the path $\mathbf{c}$ given by

$$\mathbf{c}(t) = (\sin t,\ \cos t,\ 2t)$$

is a flow line of $\mathbf{F}(x, y, z) = (y, -x, 2)$.

Example 13.7

Determine the equation for the flow lines. Hence sketch the flow lines of $\mathbf{F}(x, y) = (-y, x)$.

2.14. Differentiation Operators

Grad operator ($\boldsymbol{\nabla}$):

In $\mathbb{R}^3$,

$$\boldsymbol{\nabla} = \mathbf{i}\,\frac{\partial}{\partial x} + \mathbf{j}\,\frac{\partial}{\partial y} + \mathbf{k}\,\frac{\partial}{\partial z}.$$

By embedding $\mathbb{R}^2$ into $\mathbb{R}^3$, we can also apply $\boldsymbol{\nabla}$ to functions of two variables.

- If f is a C^1 scalar function, $\boldsymbol{\nabla} f$ is a vector field in $\mathbb{R}^3$ given by

$$\boldsymbol{\nabla} f \;=\; \mathbf{i}\,\frac{\partial f}{\partial x} + \mathbf{j}\,\frac{\partial f}{\partial y} + \mathbf{k}\,\frac{\partial f}{\partial z}.$$

Note

The ∇ symbol is called del or nabla.

Divergence of a Vector Field

If $\mathbf{F} = F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}$ is a C^1 vector field, the divergence of $\mathbf{F}$ is a scalar function given by

$$\begin{aligned}
\operatorname{div}\mathbf{F} &= \boldsymbol{\nabla}\cdot\mathbf{F} \\
&= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)\cdot(F_1, F_2, F_3) \\
&= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.
\end{aligned}$$

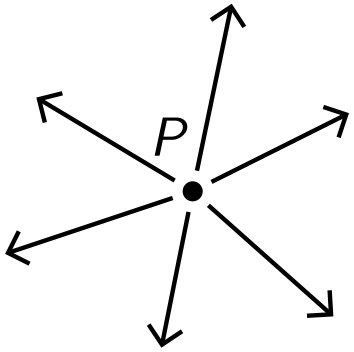
Example 14.1

If $\mathbf{F}(x, y, z) = x^2 y\,\mathbf{i} + z\,\mathbf{j} + xyz\,\mathbf{k}$, find $\boldsymbol{\nabla}\cdot\mathbf{F}$.

Physical Interpretation

If $\mathbf{F}(x, y, z)$ is the fluid velocity at position (x, y, z), then $\nabla \cdot \mathbf{F}$ measures the net transport of fluid in/out of that point. This will be made clearer when we talk about integrals.

Case 1: $\nabla \cdot \mathbf{F} > 0$
More fluid flows out than in, so the fluid is expanding or there is a source.



source at P

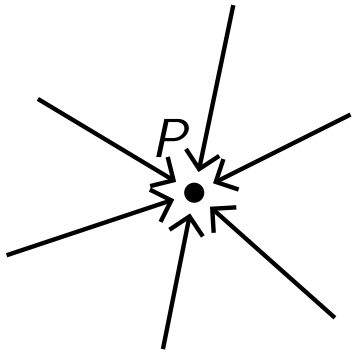
Example 14.2

If $\mathbf{F}(x, y) = x\,\mathbf{i}$, find $\nabla \cdot \mathbf{F}$.

Case 2:

$\nabla \cdot \mathbf{F} < 0$

More fluid flows in than out, so fluid is compressing or there is a sink.



sink at P

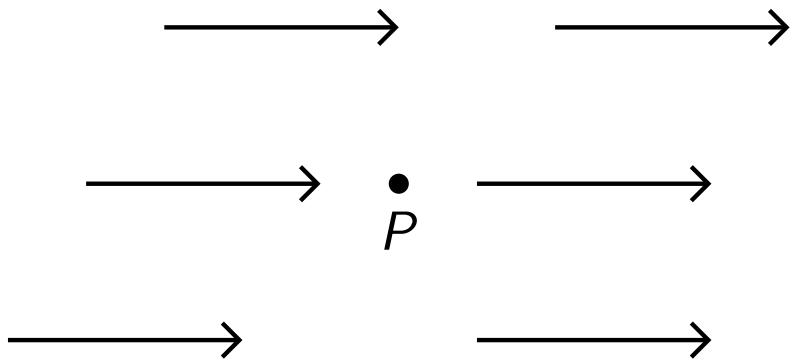
Example 14.3

If $\mathbf{F}(x, y) = -y\,\mathbf{j}$, find $\nabla \cdot \mathbf{F}$.

Case 3:

$\nabla \cdot \mathbf{F} = 0$

Rate at which fluid flows in equals rate at which fluid flows out.



no source or sink at P

If $\nabla \cdot \mathbf{F} = 0$, then $\mathbf{F}$ is an incompressible vector field.

Example 14.4

If $\mathbf{F}(x, y) = x\,\mathbf{i} - y\,\mathbf{j}$, find $\nabla \cdot \mathbf{F}$.

Curl of a Vector Field

If $\mathbf{F} = F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}$ is a C^1 vector field, the curl of $\mathbf{F}$ is a vector field in $\mathbb{R}^3$ given by

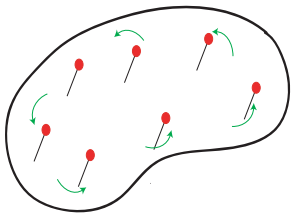
$$\begin{aligned}
\operatorname{curl}\mathbf{F} &= \boldsymbol{\nabla} \times \mathbf{F} \\
&= \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix} \\
&= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} - \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}\,.
\end{aligned}$$

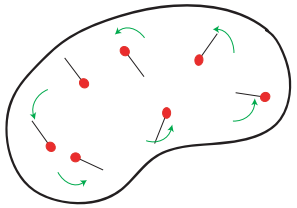
Example 14.5

If $\mathbf{F}(x, y, z) = x^2 y\,\mathbf{i} - 2xz\,\mathbf{j} + (x + y - z)\,\mathbf{k}$, find $\boldsymbol{\nabla} \times \mathbf{F}$.

Physical Interpretation

If $\mathbf{F}$ is the fluid velocity in a lake. Drop a twig into the lake. Then $\nabla \times \mathbf{F}$ measures how quickly and in what orientation the twig rotates as it moves. This will become clearer when we discuss integrals.





twig does <u>not</u> rotate as it travels

$$\Rightarrow \quad \nabla \times \mathbf{F} = \mathbf{0}$$

Direction of twig changes as it travels

$$\Rightarrow \quad \nabla \times \mathbf{F} \neq \mathbf{0}$$

If $\nabla \times \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is an irrotational vector field.

Example 14.6

Is $\mathbf{F}$, given by $\mathbf{F}(x, y, z) = x\,\mathbf{i} - y\,\mathbf{j}$, irrotational?

Example 14.7

Is $\mathbf{F}$, given by $\mathbf{F}(x, y, z) = -y\,\mathbf{i} + x\,\mathbf{j}$, irrotational?

Note

If f is a scalar function, $\nabla \cdot f$ and $\nabla \times f$ are NOT defined.

Laplacian Operator - ∇^2

In $\mathbb{R}^3$, the Laplacian operator (or just Laplace operator) is given by

$$\begin{aligned}\nabla^2 &= \boldsymbol{\nabla} \cdot \boldsymbol{\nabla} \\ &= \left(\mathbf{i}\,\frac{\partial}{\partial x} + \mathbf{j}\,\frac{\partial}{\partial y} + \mathbf{k}\,\frac{\partial}{\partial z}\right) \cdot \left(\mathbf{i}\,\frac{\partial}{\partial x} + \mathbf{j}\,\frac{\partial}{\partial y} + \mathbf{k}\,\frac{\partial}{\partial z}\right) \\ &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\end{aligned}$$

- If f is a C^2 scalar function, then $\nabla^2 f$ is a scalar function given by

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

- If $\mathbf{F} = u\,\mathbf{i} + v\,\mathbf{j} + w\,\mathbf{k}$ is a C^2 vector field where $u, v,$ and w are scalar functions of (x, y, z), then $\nabla^2\mathbf{F}$ is a vector field in $\mathbb{R}^3$ given by

$$\nabla^2\mathbf{F} = \nabla^2 u\,\mathbf{i} + \nabla^2 v\,\mathbf{j} + \nabla^2 w\,\mathbf{k}$$

Note

Some resources may denote the Laplacian by Δ.

Applications of ∇^2

Example 1

The gravitational potential V of a mass m at (x, y, z) due to a mass M at $(0, 0, 0)$ satisfies Laplace's equation

$$\nabla^2 V = 0.$$

Example 2

The temperature T of a body at (x, y, z) at time t satisfies the heat equation

$$\kappa \nabla^2 T = \frac{\partial T}{\partial t},$$

where κ is a positive constant.

Example 14.8

If $f(x, y, z) = x^2y + xy^2 + yz^2$, find $\nabla^2 f$.

Example 14.9

If $\mathbf{F}(x, y, z) = x^2 y\,\mathbf{i} + xy^2 z\,\mathbf{j} + y^3 z^2\,\mathbf{k}$, find $\nabla^2 \mathbf{F}$.

2.15. Basic Identities of Vector Calculus

Let $f, g : \mathbb{R}^3 \to \mathbb{R}$ be scalar functions. Let $\mathbf{F}$, $\mathbf{G} : \mathbb{R}^3 \to \mathbb{R}^3$ be vector fields. Then

1. $\boldsymbol{\nabla}(f+g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g$
2. $\boldsymbol{\nabla}(\beta f) = \beta \boldsymbol{\nabla} f \qquad (\beta \in \mathbb{R}$ constant$)$
3. $\boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla} g + g\boldsymbol{\nabla} f$
4. $\boldsymbol{\nabla}\left(\dfrac{f}{g}\right) = \dfrac{g\boldsymbol{\nabla} f - f\boldsymbol{\nabla} g}{g^2}$ at points (x, y, z) where $g(x, y, z) \neq 0$
5. $\boldsymbol{\nabla} \cdot (\mathbf{F} + \mathbf{G}) = \boldsymbol{\nabla} \cdot \mathbf{F} + \boldsymbol{\nabla} \cdot \mathbf{G}$
6. $\boldsymbol{\nabla} \times (\mathbf{F} + \mathbf{G}) = \boldsymbol{\nabla} \times \mathbf{F} + \boldsymbol{\nabla} \times \mathbf{G}$
7. $\boldsymbol{\nabla} \cdot (f\mathbf{F}) = f\boldsymbol{\nabla} \cdot \mathbf{F} + \mathbf{F} \cdot \boldsymbol{\nabla} f$
8. $\boldsymbol{\nabla} \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\boldsymbol{\nabla} \times \mathbf{F}) - \mathbf{F} \cdot (\boldsymbol{\nabla} \times \mathbf{G})$
9. $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{F}) = 0$
10. $\boldsymbol{\nabla} \times (f\mathbf{F}) = f\boldsymbol{\nabla} \times \mathbf{F} + \boldsymbol{\nabla} f \times \mathbf{F}$

11. $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0}$
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2\boldsymbol{\nabla} f \cdot \boldsymbol{\nabla} g$
13. $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f \times \boldsymbol{\nabla} g) = 0$
14. $\boldsymbol{\nabla} \cdot (f\boldsymbol{\nabla} g - g\boldsymbol{\nabla} f) = f\nabla^2 g - g\nabla^2 f$
15. $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{F}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{F}) - \nabla^2 \boldsymbol{F}$

Note

Identities 1–15 require f, g, $\mathbf{F}$, $\mathbf{G}$ to be suitably differentiable, either C^1 or C^2.

Example 15.1

Prove identity (11). If f is a C^2 scalar function then

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) \;=\; \mathbf{0}.$$

Scalar Potentials

Let $\mathbf{V}$ be a C^1 vector field.

If $\nabla \times \mathbf{V} = \mathbf{0}$, then $\mathbf{V}$ can be represented by the gradient of a scalar function ϕ, so

$$\mathbf{V} = \nabla\phi.$$

Note

1. ϕ is unique up to an unknown constant C.
2. The proof uses Stokes' Theorem which we will discuss later.
3. We say that $\mathbf{V}$ is a gradient field and ϕ is the scalar potential.
4. We will see later such a $\mathbf{V}$ is also called a conservative vector field.

Example 15.2

Is $\mathbf{V}$, given by

$$\mathbf{V}(x, y, z) = (2xy^2, 2x^2y + 2z^2y, 2y^2z)$$

a gradient field?

So $\mathbf{V}$ is a gradient field and we can write $\mathbf{V} = \boldsymbol{\nabla}\phi$ for some scalar function ϕ.
How do we find $\phi(x, y, z)$?

Example 15.3

Prove identity (9). If $\mathbf{F}$ is a C^2 vector field, then

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{F}) = 0$$

Vector Potentials

Let $\mathbf{V}$ be a C^1 vector field. If $\nabla \cdot \mathbf{V} = 0$, then $\mathbf{V}$ can be written as the curl of a vector field $\mathbf{F}$, so

$$\mathbf{V} = \nabla \times \mathbf{F}.$$

Note

1. $\mathbf{F}$ is not unique.
2. We say that $\mathbf{F}$ is the vector potential.
3. One can generalise the following example to produce a proof.

Example 15.4

Express the vector field $\mathbf{V}(x, y, z) = (x^2 + 1, z - 2xy, y)$, as a curl of a vector field $\mathbf{F}$.

Example 15.5

Find

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^2} \right)$$

where

$$\mathbf{r}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}\,, \;\; r = \sqrt{x^2 + y^2 + z^2}.$$

Example 15.6

Find $\nabla^2(r^2 \log(r))$ where

$$\mathbf{r}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}\,,\ r = \sqrt{x^2 + y^2 + z^2}.$$

Example 15.7

Let f and g be C^2 scalar functions. Using the vector identities, show that:

$$\boldsymbol{\nabla} \cdot (f \boldsymbol{\nabla} g - g \boldsymbol{\nabla} f) \;=\; f \nabla^2 g - g \nabla^2 f.$$